

PROJECT DESCRIPTION

Your company is specialized in the manufacturing of remote-controlled vehicles for the toy industry. Since your product are lifelike replicas of real-world vehicles, you have been selected to supply a *Project Management Package* and manage a Project that will create one new vehicle replica for the coming Holidays season. If selected, your company will have the discretion to select an aircraft (*civilian or military*), a yacht or a road vehicle (*sport, ATV or RV type*). Your company will design the selected vehicle outer body style and the inner frame to support the motor, drive, and control systems **at an approximate scale of 1/18 or (9-12 inches / 24-30 cm)**. The body can be made from steel, aluminum, or carbon fibre. The inner frame will be either aluminum or carbon fibre. You will purchase the motor and control systems from one of two Canadian suppliers. If your replica is accepted and signed off by the client, a contract will be awarded to your Company to manufacture 100,000+ replicas of your presented and accepted prototype for the next Holidays season.

You need to have a firm design choice, material's list, resources and assembly decision to show Senior Management. You are expected to start production immediately after approval of the Scope and Charter, so the replica can be delivered to your Client.

Two Canadian suppliers produce the motor, as well as the control systems, one of them you must use. Metacom Control systems (*MCS*) who are located in Surrey, British Columbia, and Paragon Technologies (*Head offices in Keflavik, Iceland*) and are located in Kingston, Ontario.

Your company has worked with MCS in the past and you have been incredibly happy with their quality and delivery records. Paragon, you have not worked with them before as they are relatively new to the industry and are promoting a new technology that they have created that might revolutionize the industry; it is not used in many applications yet and is therefore unproven. MCS has recently transferred a large portion of their production to a new plant in Ramos, Mexico in order to cut costs. Paragon's plant is located only 4 hours away from your plant here in London. For your information, Paragon's employees are in negotiation for their first labour contract since their expansion in Kingston.

Your company will need to source new assembly tooling to manufacture the toy. You have an existing manufacturing line that you could retrofit with the new assembly tooling, however it is an older technology and there have been increasing issues with production and quality. A new line would allow you to introduce new technology to increase production and quality, however there will be a learning curve for your employees and millwrights staff to learn the new line.

You can source your steel and aluminum from local suppliers, but your carbon fibre will come from Fibro Source, a plant based in Sherbrooke, Québec. You have used them in the past and have been somewhat satisfied with their quality and delivery on the last two projects that you ran. However, they have recently been bought out by a large conglomerate from Metz, France and are currently undergoing restructuring of all their departments. You have heard that there have been a large number of changes to the Canadian Management team and that a modernization of the plant including a new process equipment is underway.

Below you will find some tips (*approximate*) that will help you in budgeting your resource and material costs:



Resources:

- Project Manager = >60K – 80K<
- Team Members = >30K – 45K<
- Sundry = >8.00/hr – 15.00/hr< (i.e., General Administration, Plant maintenance, Utilities, Janitorial services, Plant Insurance, Property Taxes, Christmas Bonus, Rent, Health & Benefits per employees, Depreciation of Machinery and/or Equipment, and more)



Materials:

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|---|---------------|
| ➤ Chassis/Casting | >40K – 75K< |
| ➤ Fasteners | >5K – 15K< |
| ➤ Motor/Controller | >100K – 175K< |
| ➤ Carbon Fibre | >120K – 200K< |
| ➤ Aluminum | >50K – 115K< |
| ➤ Welding Components | >15K – 45K< |
| ➤ Circuit Boards and Peripherals | >20K – 50K< |
| ➤ Government Approval (CSA, CWS, CE, MTO) | >2K – 10K< |
| ➤ Testing | >25K – 100K< |
| ➤ Tooling | >100K – 180K< |
| ➤ Safety Equipment | >35K – 75K< |
| ➤ Packaging/Shipping | >5K – 25K< |

ATTENTION: The number given above are just a general average range. In no case we guarantee your calculation will be limited or within these parameters.